Weekly Progress Report

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Goal/Objective: Provide a final report detailing what I did after the midterm report for Group 7 of ECE494. I will give a project overview, explain my ray tracing simulations, describe the planning for our on-site mine visit, the visit itself, and the results we received from the trip.

Task 1: Explain what the project was contracted to be, what has been done on the project up until our midterm report, and how we got to that point to provide context for the remainder of the report.

Problem description

Our group was contracted by Alpha Metallurgical Resources to build an underground wireless network capable of low-latency wireless communication. The system needed to be designed for future adaptation into a fully MSHA-compliant system, and it needed to achieve sub-10ms latency, 500 kbps steady-state data rate, and 10 Mbps surge data rate capacity.

Approach

To solve the problem, we first began by educating ourselves on networking fundamentals and the OSI model. The group also read *Computer Networking: A Top-Down Approach* to build a stronger theoretical foundation.

We were able to visit the mine in the fall, which gave us a tangible understanding of the mine's constraints. From a design standpoint, we began by selecting a topology suitable for the highlevel system requirements. After the topology was determined, we selected an architecture capable of supporting it. We then chose specific components and ordered them using Alpha's grant funding.

Results

Below, I have included graphics the team developed to describe the network's architecture and topology. Due to cost constraints and the need to support long distances, we selected a daisychain topology using deterministic routing (see Figure 1 below).

At the node level, we used high-range 2.4 GHz wireless bridge modules using the 802.11 Wi-Fi protocol for wireless links. These were connected and routed using Antaira high-capacity industrial network switch modules, selected for high throughput and extremely low latency. The nodes were interconnected using PoE Cat6 cables.

To test the sensor aspect of the mine environment, we used an L3-capable industrial camera sensor to validate future integration of thermography. The node diagram is shown in Figure 2.

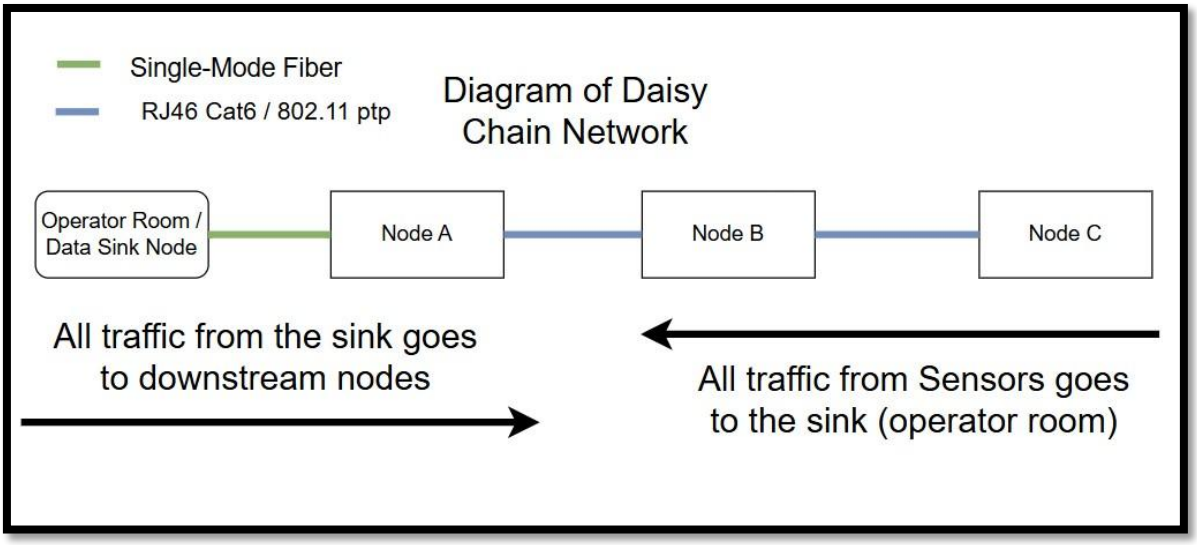


Figure 1: Daisy Chain Topology Diagram

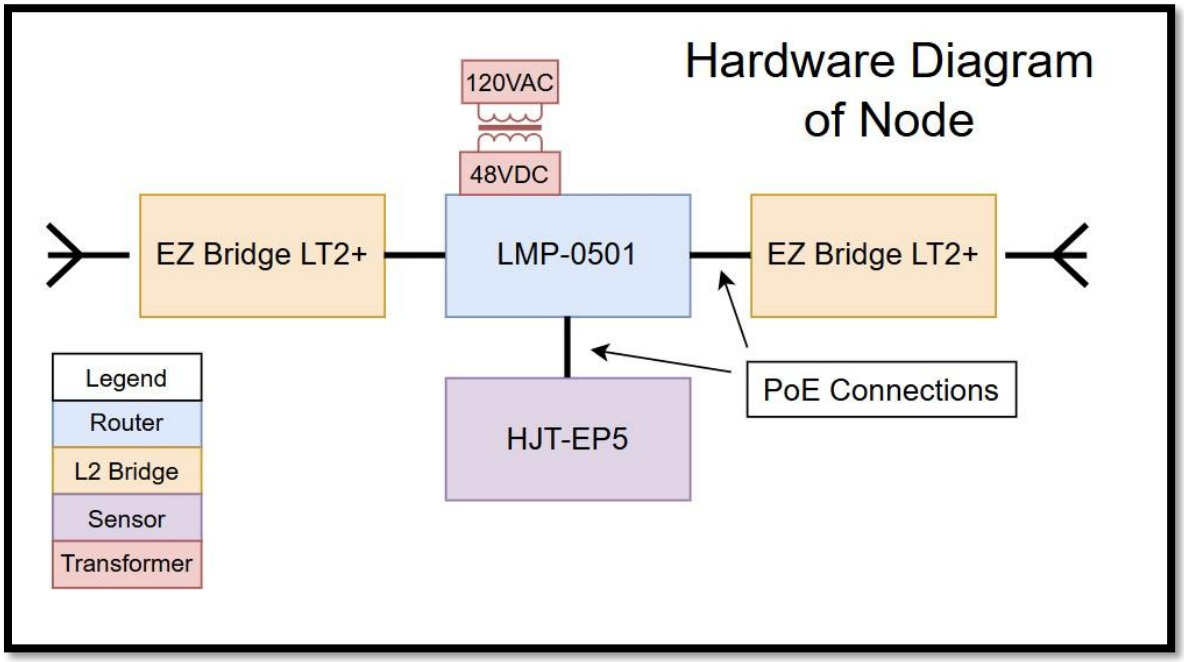


Figure 2: Hardware Diagram of Node

Task 2: Sionna RT Simulations

Problem description

To get a better picture of the RF propagation environment within the mine, as well as to provide value to Alpha's engineering team, I was instructed to complete ray tracing simulations using Sionna RT to visualize how signals propagate within the tunnels. I completed RT simulations to compare against our experimental data, so the task was to simulate environments that mimic the test sites: one outdoor scenario using UA's geometry from Google Earth, one LOS case in a tunnel, and one non-LOS case in a mine.

Approach

The first hiccup in getting the ray tracing simulation running was the toolchain required to set it up. To start, there is Sionna RT, which is an extension of Mitsuba that runs in Python as a scientific tool for optical simulations. The intended workflow for Mitsuba is to build geometry in Blender (a VFX tool), use an extension to export it as an XML file, import it into a Python script, and then process it using the Sionna RT library.

The initial problem was that none of the software tools were compatible in their most recent versions. Because of this, I had to download legacy versions of Blender, the Blender-Mitsuba addon (which was the most problematic), as well as Mitsuba itself.

After getting this running, there was a significant issue with how scenes exported from Blender as XML files are mapped into Sionna coordinates. The coordinate axes would flip and scale unpredictably. This was debugged algorithmically until a fragile fix was created using scaling adjustments and coordinate transformations.

Once the pipeline was stable, three simulations were designed and two were fully completed. The first case, shown in Figure X, was an outdoor simulation from the quad to Bevil Hall. The two points were over a kilometer apart, and only the LOS case was returned in the CIR computation. When the points were moved closer together and isotropic antennas were used, many paths were computed, which suggested the issue was related to distance as well as the geometries not being properly contained.

The second set of simulations involved an underground tunnel scenario, including both LOS and non-LOS cases. This was done to better mimic conditions expected in the underground environment. The CIRs were computed to verify validity, as well as to obtain free-space path loss values for future comparison with experimental results.

Results

The results were mixed. For the outdoor Sionna RT test, the geometry was successfully created and imported into the ray tracing simulations (see Figure 3). It provides a building map to verify correct geometry and shows a scale model of UA's main campus.

In Figure 4, you can see a 2D map with placed transmitter and receiver locations, used to ensure the simulation was set up properly. This test only yielded the LOS case, but it was an accurate

representation of free-space path loss. The result was comparable to a standard link budget computation, as well as experimentally validated using the bridge module's GUI.

The second simulation, shown in Figure 5, was conducted within a confined tunnel. Both LOS and non-LOS cases were modeled. The simulation was successfully completed for the LOS case, and its CIR plot is shown in Figure 6. The results show multiple propagation paths, with magnitudes decreasing as the paths diverge from the direct component, indicating increased bounces and attenuation.

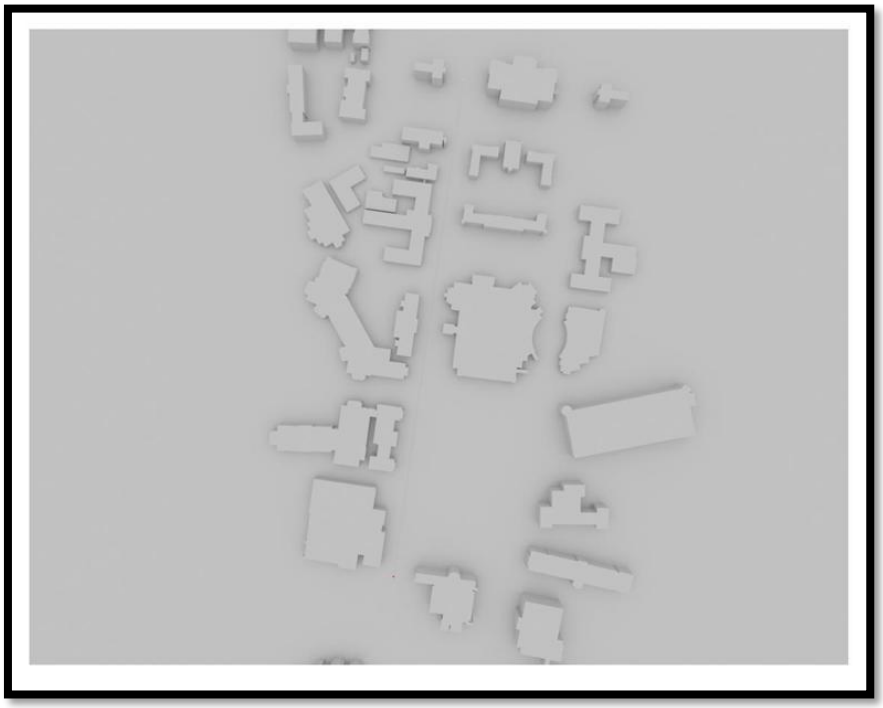


Figure 3: Outdoor Test Blender XML File

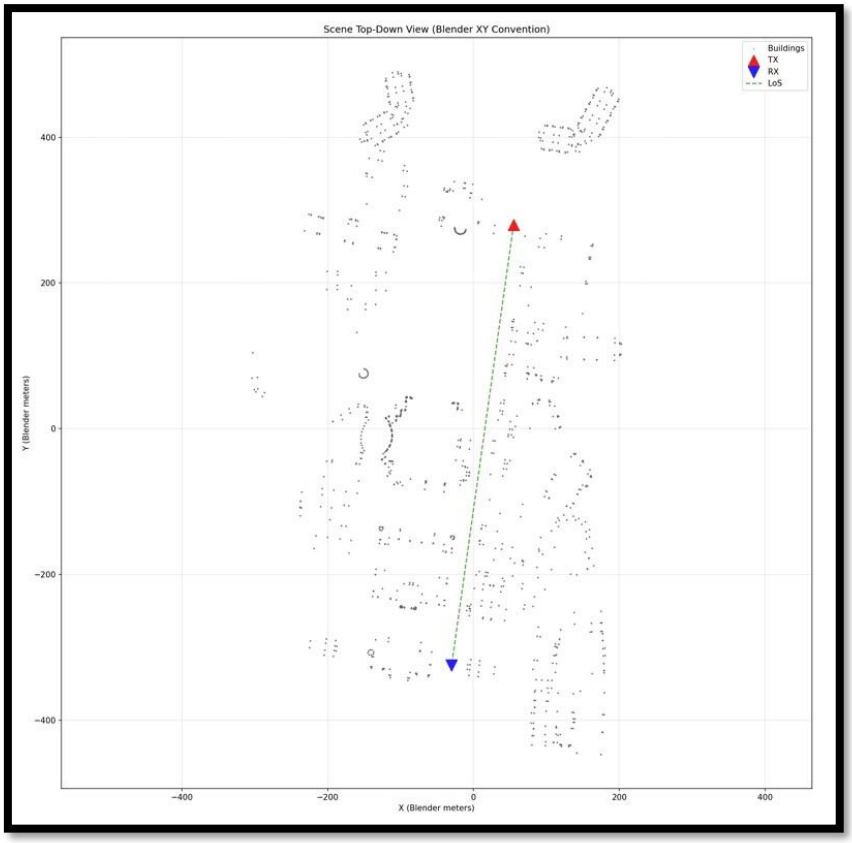


Figure 4: Outdoor Test Sionna RT 2D Test Map

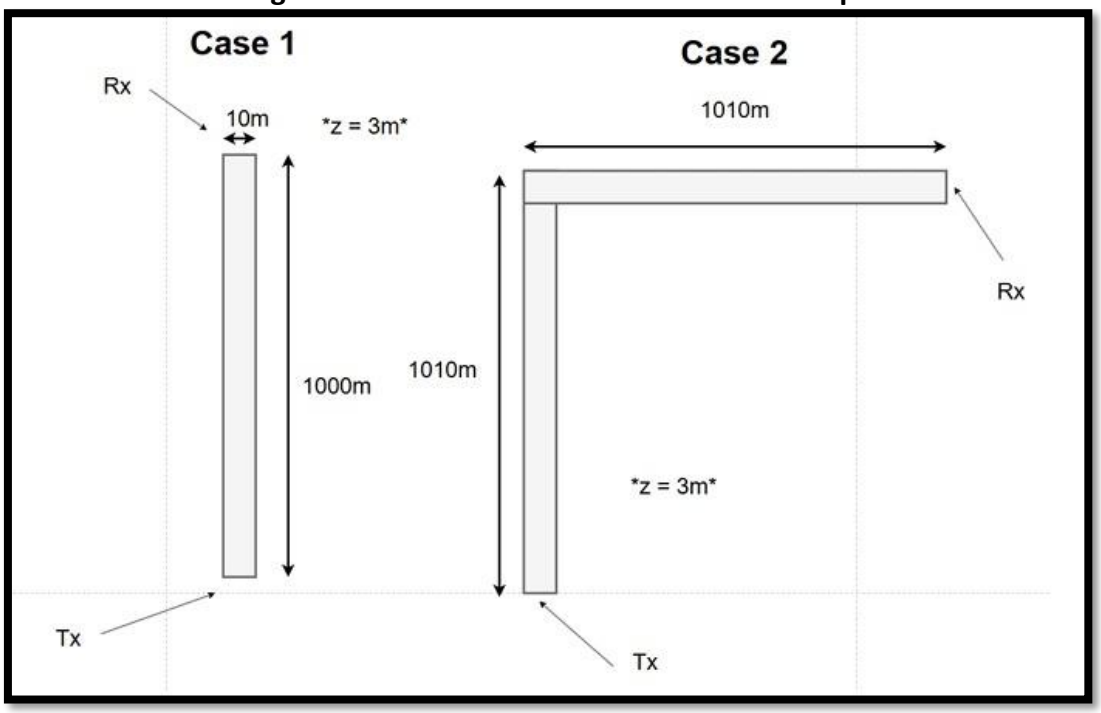


Figure 5: Tunnel RT Simulation Diagram

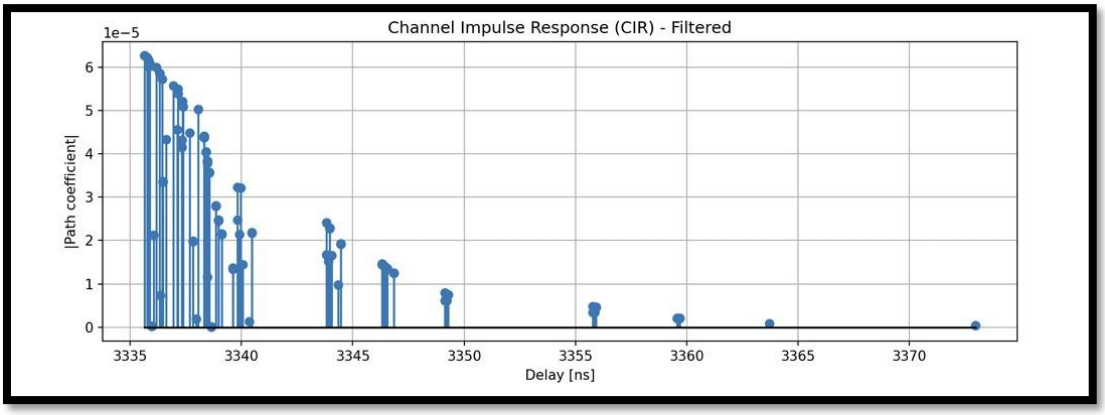


Figure 6: LOS Tunnel RT Simulation CIR Plot

Task 3: Task 3 was to prepare for and travel to Alpha's coal mine on 4/8 to successfully set up our system and collect meaningful measurements that could be presented to Alpha to demonstrate the system's performance.

Problem description

On 4/8, we planned to visit the mine, which meant we needed to create a detailed plan for the setup we wanted to implement, the tests we wanted to run, the equipment we needed to bring, and how much time could be allocated at each location, since time was scarce.

Approach

The approach was straightforward. First, we created a plan outlining what we were going to test once inside the mine and where each test would take place. Then we executed the plan onsite to collect results.

Results

I created two map diagrams for testing. Figure 7 below shows the LOS testing case in the mine, where we grouped up as Team 1 and Team 2 for each node. Figure 8 below shows the NLOS testing case.

When we went into the mine, due to power constraints, Team 2 ended up being the team assigned to the moved node for the NLOS case; otherwise, the test followed the map as shown. We also had extra time, which allowed us to go deeper into the mine and perform a long-range NLOS test of approximately 1500 feet, the longest straight shot available in the mine. Figure 9 below shows a picture of me manning the computer during testing at the far location for what became Test 3. I have also included some candid mine pictures in Figure 10, with Dr. Jeong riding a mine cart on the left and Rein setting up a node on the right.

We ended up being underground for about three hours, and Figure 11 shows a group picture taken upon completion of the underground testing.

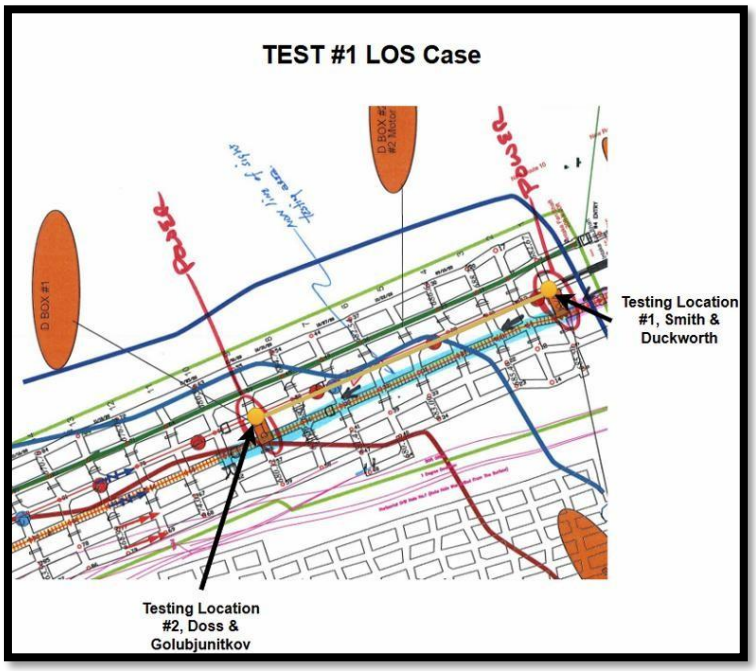


Figure 7: Map for Test Case 1 LOS

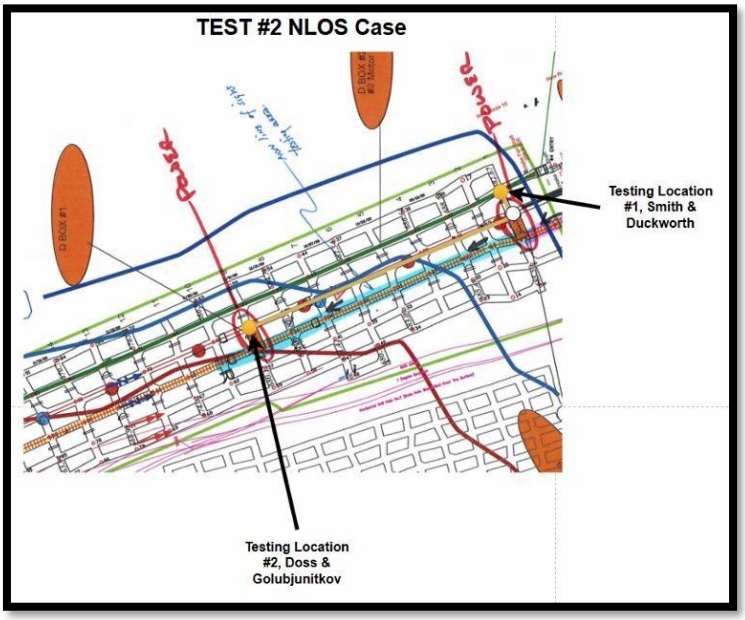


Figure 8: Map for Test Case 2 NLOS

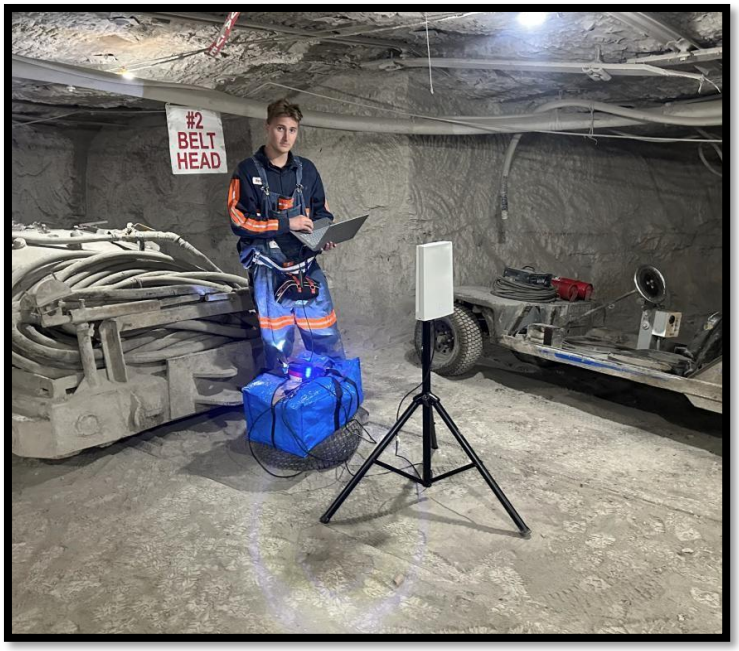


Figure 9: Node Setup in Longest NLOS Case

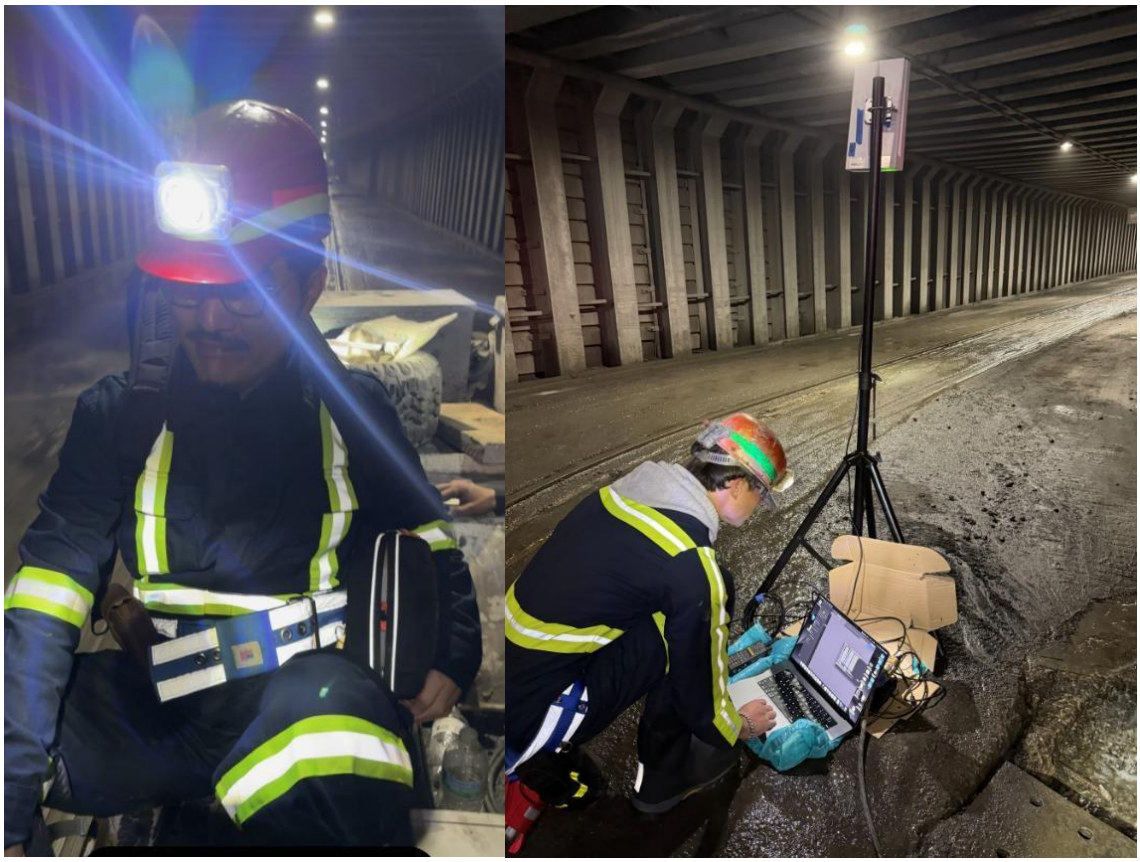


Figure 10: Candid Mine Pictures



Figure 11: Group Picture at Lynn Branch Post Testing

Task 4: Task 4 was to present and explain the data we collected from the on-site testing in the Lynn Branch mine on 4/8.

Problem description

We architected and designed a wireless network, but until it could be deployed in a real mine environment, we could not be certain how it would actually operate. On 4/8, we went to the mine to experimentally characterize the network's performance.

Approach

We ran both TCP and UDP network tests in the mine to characterize system performance in terms of latency, bandwidth, and data rate. These tests were conducted using the Iperf3 software, and the resulting data was plotted for visualization and analysis.

Results

The results were conclusive: our network clearly demonstrated performance within Alpha's criteria.

Figure 12 below shows a plot of bandwidth for each case versus test run. Across all runs and scenarios, the data rate never dropped below 90 Mbps, which is well above Alpha's required surge capacity of 10 Mbps.

Figure 13 below shows the UDP tests, which indicate latency for a node-to-node hop. Across each scenario and test, jitter averaged around 1 ms, which is well below the sub-10 ms requirement.

Overall, this information demonstrates a successful deployment that meets the desired system specifications.

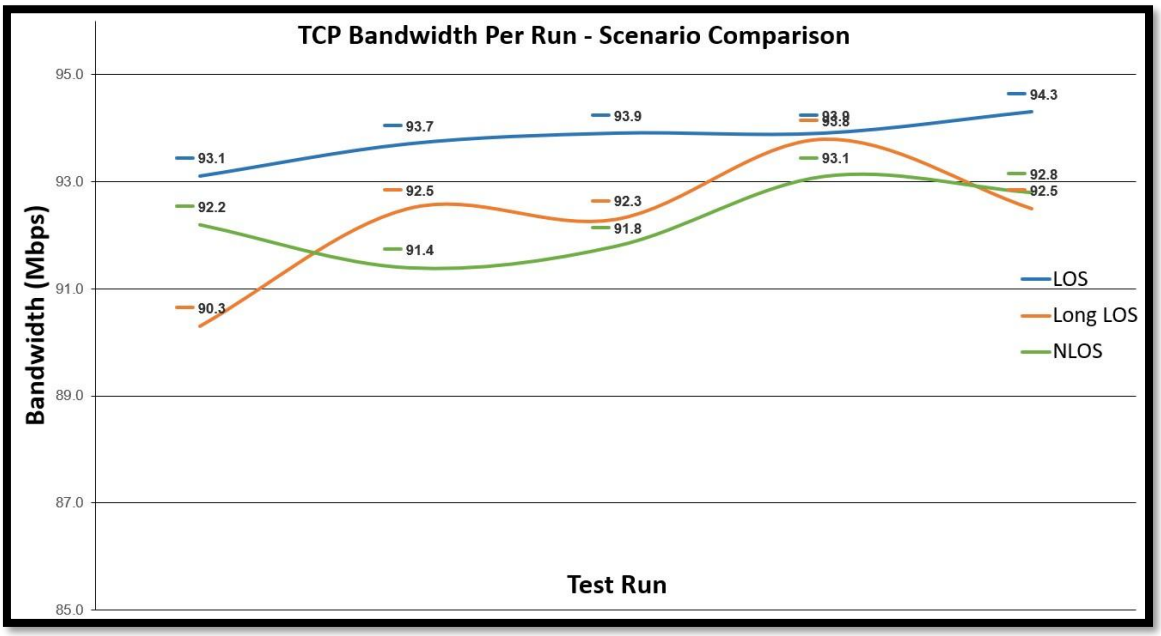


Figure 12: TCP Bandwidth vs Test Run Plot

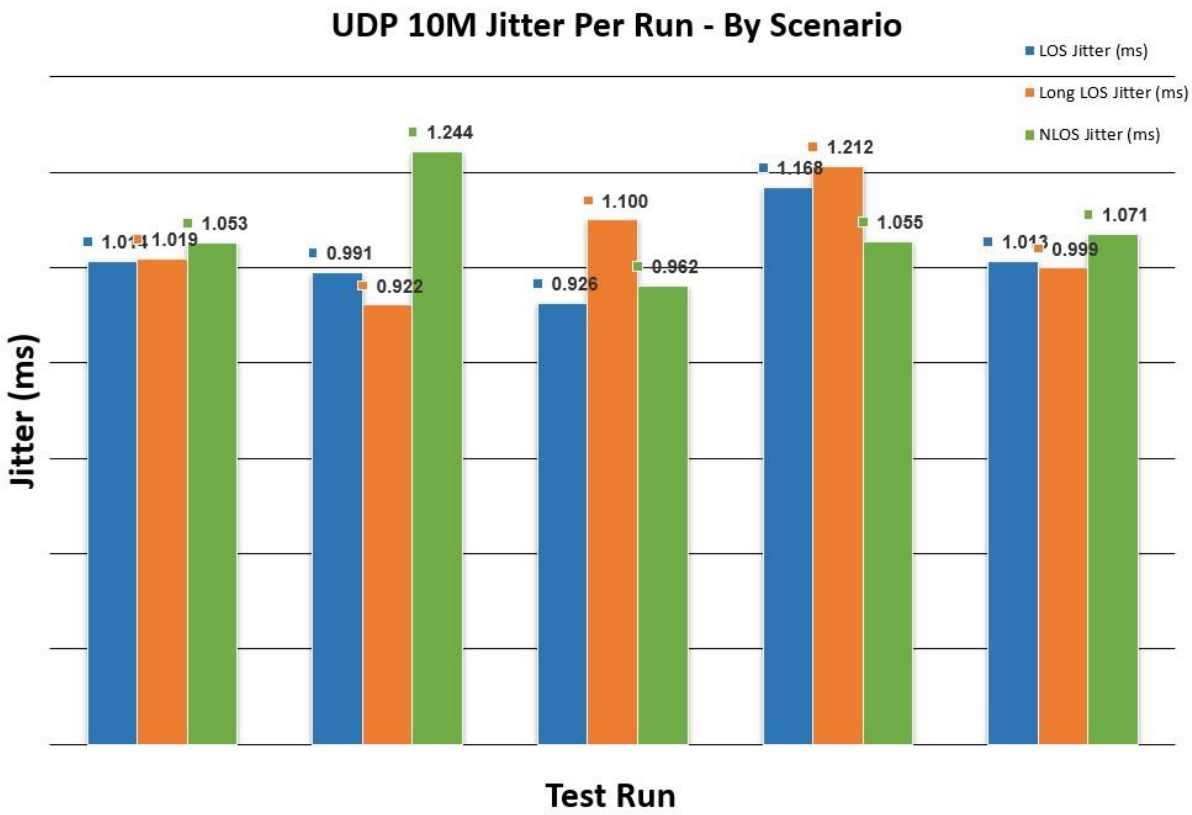


Figure 13: UDP Jitter vs Test Run Plot

References

[1] Kurose, James F, and Keith W Ross. *Computer Networking : A Top-down Approach*. 7th ed., Boston, Mass., Pearson, 2017.

[2] The Link Budget and Fade Margin, Campbell application notes